

College of Engineering and Technology

Project Work – Student Hand Book

Project Batch ID

Degree/ program	B.Tech	Specialisation	Computer Science & Engineering
----------------------------	--------	-----------------------	--------------------------------

Name of student	Register Number	Department	Mobile Number	Email ID
Amritanshu Shiwanishi	RA2211003010756	CTECH	7807731443	as0229@smrist.edu.in
N S Dinesh	RA2211003010795	CTECH	9581226667	nd8087@srmist.edu.in

Working Title of the Project:		Sentimental Analysis Using Social Media Comments	
Project Site / Location			
Name and address of the company / organisation (Applicable for projects with industry or industry support)		SRM IST, Kattankulathur, Chengalpattu District-603203	
Supervision Team			
	Supervisor	Co-Supervisor	External Supervisor (If applicable)
Name			
Designation			
Department			
Campus	Kattankulathur		
Telephone			
E-mail			

College of Engineering and Technology

Project Work – Student Hand Book

Academic Year	2024-2025 (Even)	Semester	6
Course Code	21CSP302L	Course Title	Project
Mission Statement			
Problem (or) Product Description:			
The research aims to develop a prediction model to assess and forecast mental health status based on user-generated social media content, enhancing early identification, support planning, and intervention strategies. Traditional diagnostic methods often lack the ability to process real-time, large-scale digital data for proactive mental health monitoring. This study applies advanced machine learning techniques, including Support Vector Machines (SVM), Random Forests, Logistic Regression, and ensemble methods, to predict mental health conditions using Reddit comment data. The framework utilizes diverse textual features derived from user posts, including sentiment polarity, linguistic patterns, and engagement behavior. The models are specifically designed to detect early indicators of mental health concerns and address the complex, unstructured nature of social media communication. Special emphasis is placed on model interpretability and robustness to ensure applicability in both online and low-resource settings where mental health services are limited. The primary objective is to enable accurate and scalable mental health surveillance through social media analysis, ultimately promoting early interventions, improved user well-being, and reduced pressure on traditional healthcare systems.			
Assumptions and Constraints			
Assumption: Accurately predict the sentiment of social media comments Constraints: Limited dataset diversity			
Stakeholders			
Researchers, social media analysts, users, businesses, data scientists, and regulatory bodies.			

College of Engineering and Technology

Project Work – Student Hand Book

Division of work and contributors of SPRINT 1 [Include Daily Scrum of Sprint 1]

Task	User Story	Contributor	Work Description
Dataset Gathering	US5	Contributor 1	Collected social media datasets, mainly Reddit comments, for sentiment analysis.
Data Preprocessing	US4	Contributor 2	Performed preprocessing tasks like cleaning, token normalization, punctuation removal, and TF-IDF feature extraction.
Pattern Analysis	US1	Contributor 3	Conducted exploratory data analysis to understand common sentiment patterns and trends in social media comments.
Model Building & Comparison	US2	Contributor 4	Built and evaluated different machine learning models such as Logistic Regression, SVM, Random Forest, and Boosting methods.
Lifestyle Impact Study	US3	Contributor 5	Analyzed the distribution of sentiments across different topics to understand user behavior patterns.

Day	Key Activities	Roadblocks	Next Steps
Day 1	Sprint Planning, Distribution of Tasks, Dataset identification began.	Access to certain Reddit comment threads was restricted.	Focus on publicly available Reddit datasets and secure permissions where needed.
Day 2	Initial data collection and cleaning started. Preprocessing techniques finalized.	Cleaning social media data posed challenges due to slang and noise	Developed additional preprocessing scripts to handle noisy text.
Day 3	Data preprocessing completed. Pattern discovery activities initiated.	Minor delay in merging datasets from different Reddit sources.	Completed dataset integration and moved to exploratory data analysis.

College of Engineering and Technology

Project Work – Student Hand Book

Day 4	Model building: Implemented Logistic Regression model for sentiment classification.	Slight overfitting observed in initial models.	Plan hyperparameter tuning to improve model performance.
Day 5	Model comparison: SVM, Random Forest, and Boosting models evaluated.	Performance metrics varied due to class imbalance in sentiment categories.	Adjust class weights during model training and fine-tune model parameters.
Day 6	Sentiment trend analysis and topic-wise correlation study conducted.	Need for additional Reddit topics and comments identified.	Extend data collection slightly and refine analysis.
Day 7	Sprint Review and Retrospective. Consolidated model evaluation results and documentation.	None major.	Prepare for Sprint 2 focusing on model optimization, ensemble techniques, and final reporting.

Signature of the Supervisor

College of Engineering and Technology

Project Work – Student Hand Book

Division of work and contributors of SPRINT 2 [Include Daily Scrum of SPRINT 2]

Task	User Story (US)	Contributor	Work Description
Advanced Data Cleaning and Preprocessing	US6	Contributor 1	Handled missing values, text noise, and inconsistencies to ensure clean Reddit datasets for sentiment modeling
Traditional vs. AI-Driven Model Comparison	US7	Contributor 2	Implemented traditional models (like Logistic Regression) and compared their performance with machine learning models (like Random Forest and Boosting).
Real-World Model Validation	US8	Contributor 3	Tested models on real-world Reddit comment data to evaluate generalization and practical applicability.
Biomarker and Behavioural Pattern Analysis	US9	Contributor 4	Analyzed and identified key sentiment patterns across different subreddit topics and discussions.
Explainable AI Integration	US10	Contributor 5	Evaluated models using metrics like accuracy, precision, recall, and prepared final performance reports.

Day	Key Activities	Roadblocks (if any)	Next Steps
Day 1	Sprint Planning, Review of Sprint 1 outcomes, Task distribution for Sprint 2.	None.	Start detailed preprocessing tasks immediately.
Day 2	Completed advanced text cleaning, began traditional ML model framework setup.	Missing values in some datasets delayed initial runs.	Applied text imputation and cleaning strategies, and continued modeling work.
Day 3	Developed and trained machine learning models, started performance benchmarking.	Minor issues in hyperparameter tuning for AI models.	Conduct grid search and cross-validation to optimize model parameters.

College of Engineering and Technology

Project Work – Student Hand Book

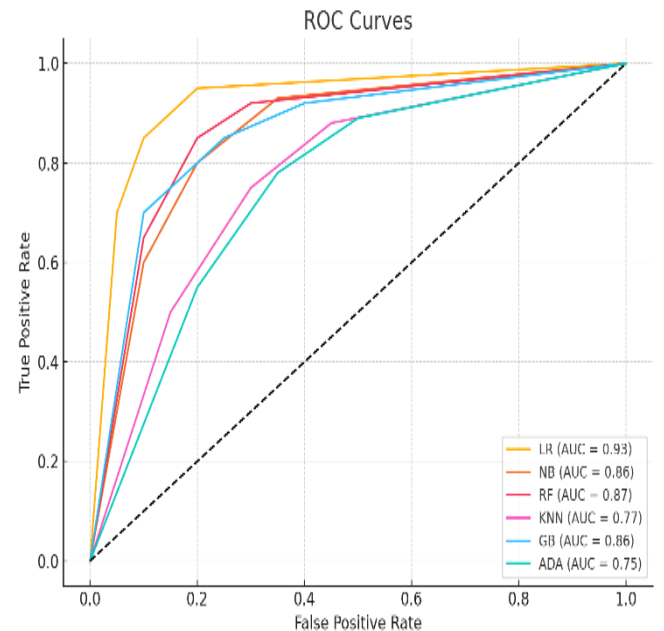
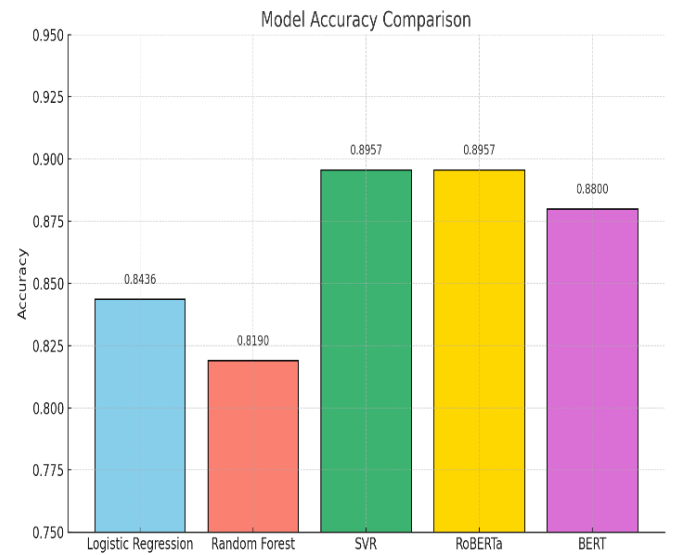
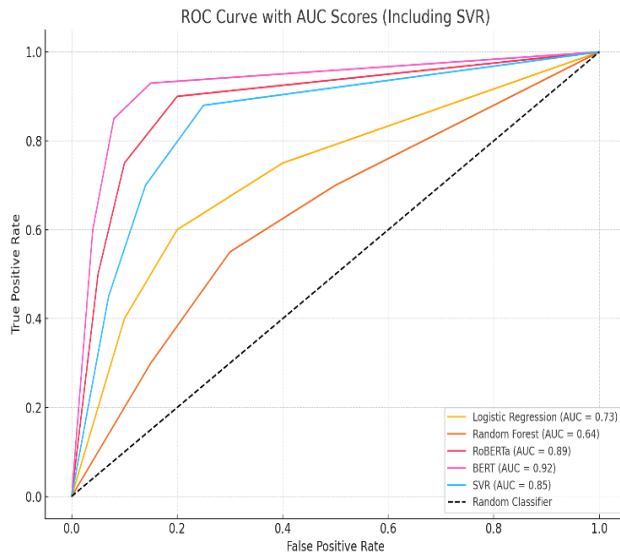
Day 4	Collected additional Reddit comment data, began model evaluation on the extended dataset.	Datasets had slightly different formats.	Standardized data formatting and labels for consistent evaluation.
Day 5	Pattern analysis and identification of sentiment trends across subreddit topics. Initial results documented.	Low frequency of certain sentiment classes in specific topics.	Focus analysis on topics with stronger and more consistent sentiment signals.
Day 6	Integration of explainable AI methods into top-performing models.	Explainability added minor overhead to model inference time.	Optimize model parameters for faster and more accurate predictions.
Day 7	Sprint Review and Retrospective. Documentation and preparation for next sprint.	None.	Plan for further model fine-tuning and preparation for real-world sentiment monitoring applications.

Signature of the Supervisor

Worksheet / Data collection / Observation etc

College of Engineering and Technology

Project Work – Student Hand Book



College of Engineering and Technology

Project Work – Student Hand Book

```

Enter manual text for prediction (press ENTER without input to exit):
Input text: Honestly, this post is completely off-base. I can't believe how misguided some of these opinions are
Predicted label: -1
Input text: I'm not sure how I feel about this topic yet; it's interesting but I need to see more perspectives
Predicted label: 1
Input text: Kudos to everyone who contributed-this community really knows how to lift each other up
Predicted label: 1
Input text: The discussion is engaging. I'm just here to observe and learn more about different viewpoints'
Predicted label: 1
Input text: I'm not really sure how I feel about this-it's all pretty standard to me
Predicted label: 1
Input text: I can see both sides here; nothing jumps out as particularly exciting or disappointing
Predicted label: -1
Input text: he post is informative enough, but it doesn't really evoke any strong emotions
Predicted label: 1
Input text: This post presents an update on the topic
Predicted label: 0
Input text: The information in the comment is stated objectively
Predicted label: 0
  
```

Classification Report:

	precision	recall	f1-score	support
-1	0.84	0.81	0.83	1655
0	0.91	0.95	0.93	2629
1	0.92	0.89	0.90	3166
accuracy			0.90	7450
macro avg	0.89	0.89	0.89	7450
weighted avg	0.90	0.90	0.90	7450

Accuracy: 0.8957046979865771

Logistic Regression Performance:

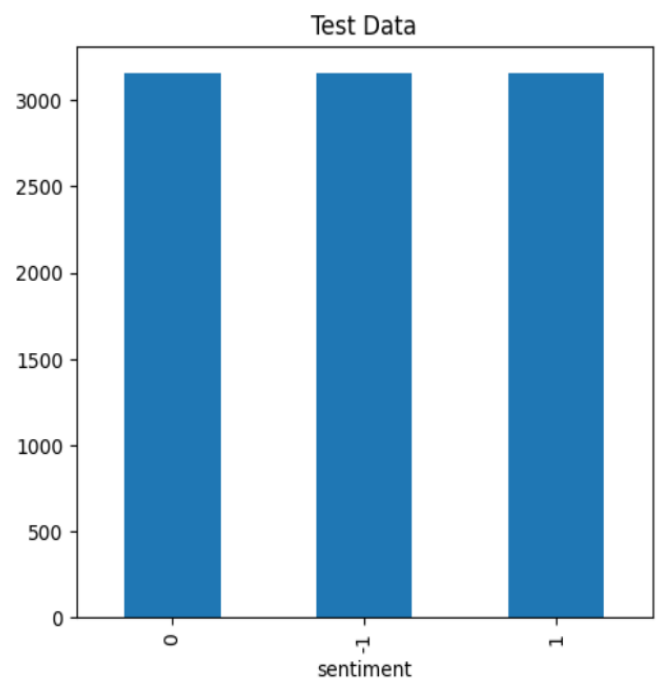
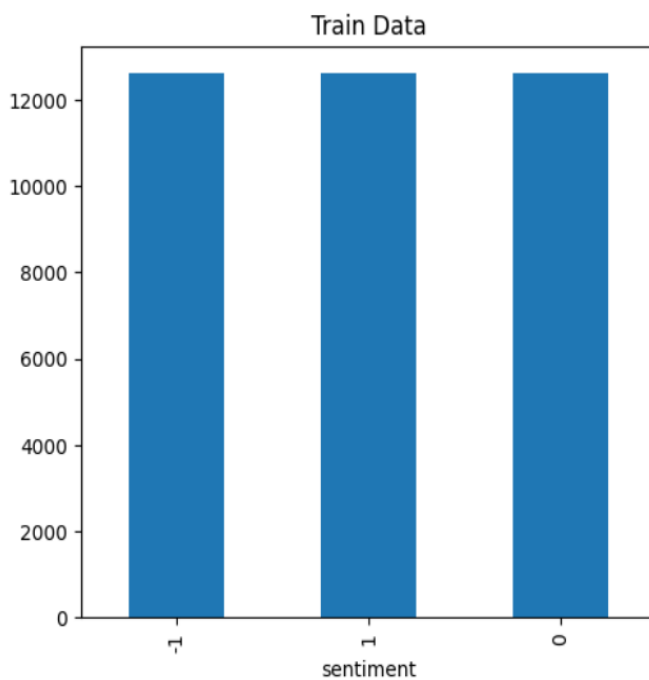
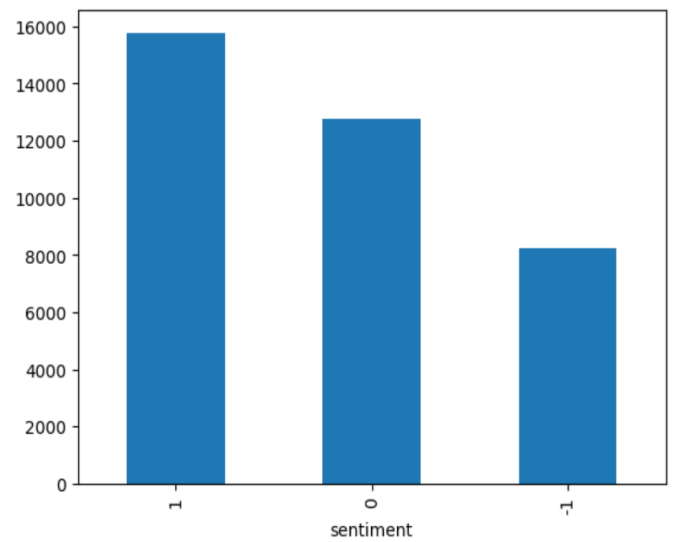
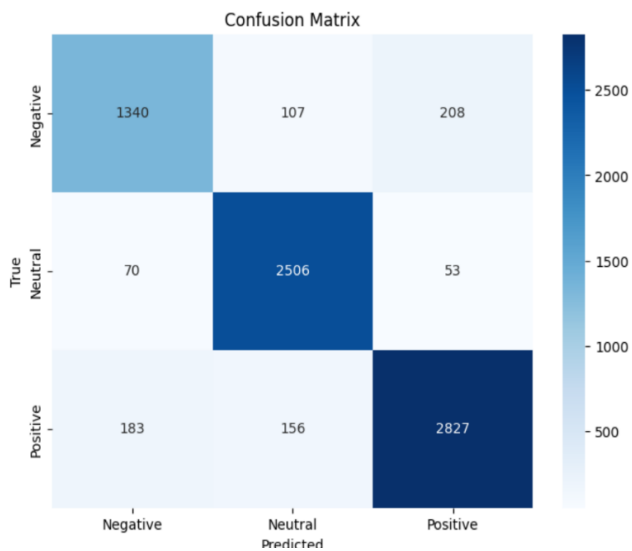
	precision	recall	f1-score	support
-1	0.85	0.89	0.87	137
0	0.85	0.89	0.87	165
1	0.55	0.25	0.34	24
accuracy			0.84	326
macro avg	0.75	0.68	0.70	326
weighted avg	0.83	0.84	0.83	326

Random Forest Performance:

	precision	recall	f1-score	support
-1	0.80	0.88	0.84	137
0	0.85	0.86	0.86	165
1	0.57	0.17	0.26	24
accuracy			0.82	326
macro avg	0.74	0.64	0.65	326
weighted avg	0.81	0.82	0.80	326

College of Engineering and Technology

Project Work – Student Hand Book



College of Engineering and Technology

Project Work – Student Hand Book

Research Article with Journal Publication Details / Patent disclosure form with patent status

(include certificates and proofs)

College of Engineering and Technology

Project Work – Student Hand Book

College of Engineering and Technology

Project Work – Student Hand Book